Problem Set 2.2 (pp. 72–73)

Find a general sol: of the following DEq:s

Q.2. $y'' - 9y = 0$

Sol: Here the characteristic (Auxiliary) eq: is

$$\lambda^2 - 9 = 0$$

$$\Rightarrow \quad \lambda^2 = 9$$

$$\Rightarrow \quad \lambda = \pm 3$$

$\therefore$ A basis is e^{3x}, e^{-3x}

Hence the general sol: is

$$y = c_1 e^{3x} + c_2 e^{-3x}$$

Q.4 $y'' + 4y' = 0$

Sol: Here the characteristic eq: is

$$\lambda^2 + 4\lambda = 0$$

$$\Rightarrow \quad \lambda(\lambda + 4) = 0$$

$$\Rightarrow \quad \lambda = 0, \ \lambda = -4$$

$\therefore$ A basis for sol: is e^{0x}, e^{-4x} or $1, e^{-4x}$

(88) Hence the general sol: is

$$y = c_1 + c_2 e^{-4x}$$

Q.6. $\quad y'' - 2y' + 0.75y = 0$

Sol: Here the characteristic eq: is

$$\lambda^2 - 2\lambda + 0.75 = 0$$

Solving,

$$\lambda = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(0.75)}}{2(1)}$$

$$= \frac{2 \pm \sqrt{4 - 3.00}}{2}$$

$$= \frac{2 \pm 1}{2}$$

$$= \frac{3}{2}, \frac{1}{2}$$

∴ A basis for the sol: is $e^{\frac{3}{2}x}, e^{\frac{1}{2}x}$

Hence the general sol: is

$$y = c_1 e^{\frac{3}{2}x} + c_2 e^{\frac{1}{2}x}$$

Solve the following IVPs.

Q.18 $y'' - 4y' + 4y = 0, \; y(0) = 0, \; y'(0) = -3$

Sol: Here the characteristic eqn is

$$\lambda^2 - 4\lambda + 4 = 0$$

or $$(\lambda - 2)^2 = 0$$

or $$\lambda = 2, 2$$

$\therefore$ A basis for the sol. is $e^{2x}, \; xe^{2x}$.

Hence the general sol. is

$$y = c_1 e^{2x} + c_2 x e^{2x}$$

or $$y = (c_1 + c_2 x) e^{2x} \quad \text{--- (1)}$$

Also $$y' = c_2 e^{2x} + 2(c_1 + c_2 x) e^{2x}$$

or $$y' = 2c_1 e^{2x} + c_2(1 + 2x) e^{2x} \quad \text{--- (2)}$$

From (1), $y = 0$ when $x = 0$

$\therefore$ $0 = c_1$

From (2), $y' = -3$ when $x = 0$.

$\therefore$ $-3 = 2c_1 + c_2$. But $c_1 = 0$

$\therefore$ $c_2 = -3$

(90) Hence the sol: of the IVP is

$$y = -3xe^{7x}$$

Q.20. $y'' + 3.7y' = 0,\ y(-2) = 4,\ y'(-2) = 0$

Sol: Here the characteristic eq: is

$$\lambda^2 + 3.7\lambda = 0$$

$\therefore$ $$\lambda(\lambda + 3.7) = 0$$

or $$\lambda = 0,\ \lambda = -3.7$$

$\therefore$ A basis for the sol: is $e^{0x}, e^{-3.7x}$ or $1, \dots$

Thus, the general sol: is

$$y = c_1 + c_2 e^{-(3.7)x} \quad \text{——— (1)}$$

Also,

$$y' = (-3.7)c_2 e^{-(3.7)x} \quad \text{——— (2)}$$

Now initially, $y = 4$ when $x = -2$

$\therefore$ (1) gives

$$4 = c_1 + c_2 e^{7.4} \quad \text{——— (3)}$$

Again, $y' = 0$ when $x = -2$

$\therefore$ (2) gives

$$0 = (-3.7)c_2 e^{7.4} \quad \text{——— (4)}$$

$$\Longrightarrow c_2 = 0$$

$\therefore$ From (3), $c_1 = 4$

Hence the sol: of the IVP is

$$y = 4$$

Q.22 $y'' - 25y = 0, \quad y(0) = 0, \quad y'(0) = 10$

Sol: Characteristic Eq: is

$$\lambda^2 - 25 = 0$$

or $\lambda^2 = 25$

or $\lambda = \pm 5$

∴ A basis for the sol: is e^{5x}, e^{-5x}

Thus the general sol: is

$$y = c_1 e^{5x} + c_2 e^{-5x} \quad \text{——— (1)}$$

Also, $y' = 5c_1 e^{5x} - 5c_2 e^{-5x}$ ——— (2)

From $y(0) = 0$, we get from (1)

$$0 = c_1 + c_2 \quad \text{——— (3)}$$

and from $y'(0) = 10$, we get from (2)

$$10 = 5c_1 - 5c_2$$

or $2 = c_1 - c_2$ ——— (4)

or $\{(-8-6i)+(-2+6i)+10\}e^{(-1+3i)x}$ (95)

$+\{(-8+6i)+(-2-6i)+10\}e^{(-1-3i)x}$

$= 0e^{(-1+3i)x} + 0e^{(-1-3i)x}$

$= 0 =$ R.H.S.

$\therefore$ The given function satisfies the eq: Hence it is a sol: of the eq:

Again, $\because y'' + 2y' + 10y = 0$

$\therefore$ Here the characteristic eq: is

$$\lambda^2 + 2\lambda + 10 = 0$$

or $$\lambda = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(10)}}{2(1)}$$

$$= \frac{-2 \pm \sqrt{4-40}}{2}$$

$$= \frac{-2 \pm \sqrt{-36}}{2} = \frac{-2 \pm 6i}{2} = -1 \pm 3i$$

$\therefore$ A basis is $e^{(-1+3i)x}, e^{(-1-3i)x}$

Hence the general sol: is

$$y = c_1 e^{(-1+3i)x} + c_2 e^{(-1-3i)x}.$$

(94) # Problem Set 2.3 (pp. 76-77)

Verify that the following functions are (solutions)
of the given DE and obtain from them
real valued general sol: of the form (...)

Q.2 $y = c_1 e^{(-1+3i)x} + c_2 e^{(-1-3i)x}$,

$$y'' + 2y' + 10y = 0$$

Sol: $\because$ $y = c_1 e^{(-1+3i)x} + c_2 e^{(-1-3i)}$

$\therefore$ $y' = (-1+3i) c_1 e^{(-1+3i)x} + (-1-3i) c_2 e^{(-1-...}$

and $y'' = (-1+3i)^2 c_1 e^{(-1+3i)x} + (-1-3i)^2 c_2 e^{(-...}$

Substituting these values in the L.H.S. of the given

$(-1+3i)^2 c_1 e^{(-1+3i)x} + (-1-3i)^2 c_2 e^{(-1-3i)x}$
$+ 2\{(-1+3i) c_1 e^{(-1+3i)x} + (-1-3i) c_2 e^{(-1-3i)x}\}$
$+ 10\{c_1 e^{(-1+3i)x} + c_2 e^{(-1-3i)x}\}$

or $\{(-1+3i)^2 + 2(-1+3i) + 10\} c_1 e^{(-1+3i)x}$
$+ \{(-1-3i)^2 + 2(-1-3i) + 10\} c_2 e^{(-1-3i)x}$

Q.8 $y'' - 8y' + 16y = 0$

Sol: Here the characteristic eq: is

$$\lambda^2 - 8\lambda + 16 = 0$$

Solving,
$$\lambda = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(16)}}{2(1)}$$

or $$\lambda = \frac{8 \pm \sqrt{64 - 64}}{2} = \frac{-8 \pm \sqrt{0}}{2}$$

It has two $\frac{-8}{2} = -4, -4$ equal (repeated) roots

∴ The given eq: corresponds to case II.

Hence the general sol: is

$$y = (c_1 + c_2 x) e^{-4x}.$$

Q.10 $y'' + y' + 0.25y = 0$

Sol: Here the characteristic eq: is

$$\lambda^2 + \lambda + 0.25 = 0$$

Solving
$$\lambda = \frac{-1 \pm \sqrt{(1)^2 - 4(1)(0.25)}}{2}$$

$$= \frac{-1 \pm \sqrt{1 - 1}}{2} = -\frac{1}{2}, -\frac{1}{2}$$

Its roots are real and distinct

$\therefore$ It corresponds to Case I.

A basis ~~of the~~ sol: is $e^{\frac{1}{2}x}, e^{-\frac{1}{4}x}$

Hence the general sol: is

$$y = c_1 e^{\frac{1}{2}x} + c_2 e^{-\frac{1}{4}x}$$

Q. 14 $2y'' + 10y' + 25y = 0$.

Sol: Here the characteristic eq: is

$$2\lambda^2 + 10\lambda + 25 = 0$$

Solving,

$$\lambda = \frac{-10 \pm \sqrt{(10)^2 - 4(2)(25)}}{2(2)}$$

$$= \frac{-10 \pm \sqrt{100 - 200}}{4}$$

$$= \frac{-10 \pm \sqrt{-100}}{4}$$

$$= \frac{-10 \pm 10i}{4}$$

$$= \frac{1}{2}(-5 \pm 5i)$$

$\therefore$ The roots are complex conjugates, the eq: corresponds to Case III.

(126) Solve the following IVPs

Q.14. $x^2y'' - 4xy' + 4y = 0,$
$y(1) = 4, \quad y'(1) = 13.$

Sol: The aux. eq. is

$$m(m-1) - 4m + 4 = 0$$

or $$m^2 - m - 4m + 4 = 0$$

or $$m^2 - 5m + 4 = 0$$

or $$(m-4)(m-1) = 0$$

or $$m = 4, 1$$

The roots being real and distinct, a basis for the sol. is

$$x^4, x$$

$\therefore$ The general sol. is

$$y = c_1 x^4 + c_2 x \quad \text{—— (1)}$$

Also, $$y' = 4c_1 x^3 + c_2 \quad \text{—— (2)}$$

Using the initial conditions,

$$4 = c_1 + c_2 \quad \text{and} \quad 13 = 4c_1 + c_2$$

By subtraction, $3c_1 = 9$ or $c_1 = 3$ $\therefore c_2 = 1$. (12)

Hence the sol: of the IVP is

$$y = 3x^4 + x$$

Q.16 $(x^2D^2 - 5xD + 8)y = 0$,

$y(1) = 5$, $y'(1) = 18$.

Sol: The aux: eq: is

$$m(m-1) - 5m + 8 = 0$$

or $$m^2 - m - 5m + 8 = 0$$

or $$m^2 - 6m + 8 = 0$$

or $$(m-4)(m-2) = 0$$

or $$m = 4, 2$$

The roots being real and distinct, a basis for the sol: is x^4, x^2

Hence the general sol; is

$$y = c_1 x^4 + c_2 x^2 \text{ ——— (1)}$$

$$y' = 4c_1 x^3 + 2c_2 x \text{ ——— (2)}$$

(128) Using the initial conditions

$$5 = c_1 + c_2 \quad \text{———} (3)$$
$$18 = 4c_1 + 2c_2 \quad \text{———} (4)$$

Multiplying (3) by 2 and subtracting fr

$$8 = 2c_1 \text{ or } c_1 = 4$$

$\therefore$ (3) gives $c_2 = 1$

From (1), putting the values of c_1, c_2 the sol: of the IVP is

$$y = 4x^4 + x^2.$$

Q. 18 $10x^2y'' + 46xy' + 32.4y =$ ...
$y(1) = 0, \; y'(1) = 2.$

Sol: Here the aux: eq: is

$$10m(m-1) + 46m + 32.4 =$$
or $$10m^2 - 10m + 46m + 32.4$$